

TEXT-TO-MUSIC-TO-TEXT-TO-MUSIC: A RESEARCH STUDY TO EXPLORE AI'S CAPABILITY IN PRESERVING GENRE AND INSTRUMENTAL COHERENCE THROUGH ITERATIVE GENERATIONS

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ABSTRACT

This project aims to investigate the capabilities of artificial intelligence (AI) in music generation and audio captioning, focusing on two primary objectives: evaluating the ability of AI systems to maintain semantic and musical coherence across multiple iterations and analyzing the instruments variations in musical and textual content through iterative processes. We employ a novel iterative process to evaluate the stability and accuracy of genre recognition and instrument coherence in AI-generated music. By investigating these aspects, this project aims to advance understanding of AI's potential in creative fields, highlighting its strengths and identifying areas for improvement.

Index Terms— MusicGen, Demucs, LP-Music Caps, generative music, stem demuxing, music captioning, genre recognition

1. INTRODUCTION

The integration of artificial intelligence (AI) in music generation represents a significant leap in creative technology, tracing its origins back to the mid-20th century with early rule-based systems [1].

These systems were limited to predefined patterns, producing simplistic compositions. The field has since evolved, with notable advancements driven by Generative Adversarial Networks (GANs) and Transformer models. GANs, introduced by Goodfellow et al. (2014), facilitate the creation of complex and realistic musical pieces through a dual-network training process: one generating music and the other evaluating it [2].

Transformers, popularized by Vaswani et al. (2017), have been adapted for music generation, enabling the production of longer and more contextually aware compositions [3].

Recent discoveries in AI's ability to generate music and perform audio captioning have further propelled the field. For example, the application of convolutional neural networks (CNNs) and recurrent neural networks (RNNs) in audio captioning has demonstrated significant improvements in translating auditory information into descriptive text.

These techniques, initially inspired by image captioning methods, have been adapted to handle the unique challenges of audio data, such as using spectrograms as input for CNNs that then feed into RNNs or Transformer models [4, 5].

Our project builds on these advancements, utilizing state-of-the-art models such as MusicGen for text-to-music conversion, Demucs for music source separation, and LP-Music Caps for audio captioning. A novel iterative process has been developed to assess

the stability and precision of genre recognition in AI-generated music.

The questions we want to answer are the following: if I generate music using a prompt (e.g. “A traditional blues track featuring a slow, soulful tempo...”), i demux a stem (e.g. the bass track), and feed it into the music captioning model, *would I obtain a blues bass description?* Also, if I re-generate music based on the last caption, *would I obtain a bass only track?*

This project underscores the importance of continued research in AI-driven music generation, improving coherence and adaptability across different musical styles. It aims to contribute to the growing body of knowledge and provide insights for developing more sophisticated and reliable AI systems in the creative industry.

2. METHODOLOGY

The following state-of-the-art models are used in the implementation for music generation, demuxing, captioning, genre recognition. They are subsequently integrated to form the core of project: an original iterative process.

2.1. Music Generation with MusicGen

MusicGen[6] is a state-of-the-art AI model developed by Meta, designed to generate high-quality music from textual descriptions. The model leverages a single-stage auto-regressive Transformer architecture trained over a 32kHz EnCodec tokenizer with 4 codebooks sampled at 50 Hz.

MusicGen employs multi-head attention mechanisms to process and generate music. The attention mechanism is mathematically defined as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (1)$$

where Q , K , and V are the query, key, and value matrices derived from the input text, and d_k is the dimension of the keys and queries. This mechanism allows the model to focus on different parts of the input text simultaneously, aligning the generated music closely with the semantic content of the textual prompt.

The input text is first converted into dense vector representations through an embedding layer, transforming text tokens into a format suitable for the Transformer model. The output of the Transformer model is then mapped to musical elements through a softmax layer, which provides a probability distribution over possible musical outputs. These outputs are decoded into audio using a pre-trained EnCodec model.

The MusicGen model is trained using a cross-entropy loss function, which measures the difference between the predicted probability distributions and the actual distribution of musical elements. Additionally, the model uses evaluation metrics such as the Frechet Audio Distance (FAD)[7] and Kullback-Leibler Divergence (KLD)[8] to ensure high-quality music generation.

2.2. Demuxing with Demucs

Demucs, a U-net architecture featuring an encoder and decoder, utilizes a series of convolutional layers that progressively downsample the input waveform and then upsample it to produce separate waveforms for each source. This model is particularly effective for separating drums, bass, and vocals from the rest of the accompaniment [9, 10].

For this project, we utilized the parameter setting `model=mdx_extra`, which includes pre-trained weights to enhance the model's capability in source separation. Another critical parameter, `segment`, is set to 12. This setting determines the size of the audio chunks processed by the model, with larger values necessitating more memory but potentially improving the quality of the separation by providing better context.

The effectiveness of Demucs in separating audio sources is quantified using the spectral convergence loss, which minimizes the difference between the magnitudes of the predicted and actual spectrograms. The spectral convergence loss is defined as follows:

$$L_{\text{spectral}} = \frac{\| |S| - |\hat{S}| \|_F}{\| |S| \|_F} \quad (2)$$

where $|S|$ and $|\hat{S}|$ represent the magnitudes of the actual and predicted spectrograms, respectively, and $\| \cdot \|_F$ denotes the Frobenius norm. This loss function is crucial for ensuring that the separated audio closely matches the original components in the frequency domain, leading to more accurate and musically coherent outputs.

2.3. Music Captioning with LP-Music Caps

LP-Music Caps (LP-MusicCaps: LLM-Based Pseudo Music Captioning) [11] is an advanced AI model designed to create textual descriptions of audio content. It leverages a large pre-trained language model (LLM), such as GPT-3.5, fine-tuned specifically for music captioning tasks. The architecture of LP-Music Caps is based on a transformer framework, enabling it to capture complex temporal dynamics and patterns within musical compositions.

The model processes audio features such as melody, harmony, rhythm, and timbre to produce detailed and contextually relevant captions.

The caption generation process involves transforming the input audio into feature vectors that the model can interpret. These features are then processed by the transformer model to generate textual descriptions. The sequence-to-sequence approach of the transformer allows it to predict the next word in the caption based on the preceding words and the audio features.

The LP-Music Caps model utilizes a cross-entropy loss function during training to measure the difference between predicted and actual probability distributions of musical elements:

$$L = - \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (3)$$

where N is the number of training examples, y_i is the actual label in one-hot encoded form, and $\hat{y}_i$ is the predicted probability of the label.

LP-Music Caps employs both zero-shot and transfer learning scenarios for evaluation. Zero-shot learning involves generating captions for audio content not seen during training, while transfer learning fine-tunes the model on a smaller, task-specific dataset to enhance performance.

2.4. Genre recognition

For genre recognition in this project, we use a convolutional neural network (CNN) previously employed in a parallel research study at Polimi[12]. This CNN, composed of various specialized layers, transforms and reduces the dimensions of the embeddings [13, 14, 15]. The architecture includes input, convolutional, max-pooling, batch normalization, dropout, flatten, and dense layers, culminating in an output layer with 10 units representing different genres.

This CNN was trained using the public dataset GTZAN[16] (80% training, 20% validation, embeddings randomly selected), which is organized into 10 categories: Blues, Classical, Country, Disco, Hip hop, Jazz, Metal, Pop, Reggae and Rock.

Each category has 100 songs, resulting in a total of 1000 songs.

3. IMPLEMENTATION

The iterative process involved running the system three times, each with a different initial prompt and a different target genre (i.e., blues, classical, and reggae). The workflow for each genre prompt, as illustrated in Figure 1, is described below.

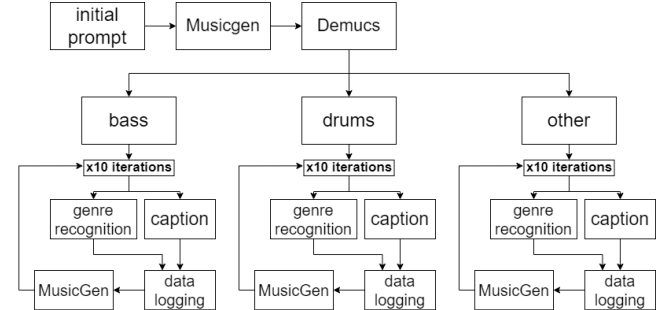


Figure 1: Project implementation diagram

Starting from an initial prompt, MusicGen generated a music piece which was then separated into its constituent components (bass, drums, and other) using Demucs.

Each of these components underwent ten iterations, during which the following steps were performed:

- **Captioning:** The generated music piece was described using LP-Music Caps to produce a textual caption.
- **Regeneration:** The caption was fed back into MusicGen to regenerate the music piece.
- **Genre Recognition:** The regenerated music was classified using a convolutional neural network (CNN) trained on the GTZAN dataset to recognize its genre.
- **Data Logging:** Throughout the process, data on genre recognition confidence scores and the generated captions were logged for analysis.

The ten iterations aimed to evaluate the stability and accuracy of genre recognition and instrument coherence in AI-generated music. For each prompt (i.e. blues, classical and reggae) and each iteration the confidence scores from the original genre served as a benchmark against which the predicted genres and their respective confidence scores were compared.

4. RESULTS

For each prompt and each iteration, we collected the confidence score associated with the target genre (i.e., the genre specified alongside the initial prompt) to measure its deviation. Additionally, we gathered data on the recognized genre and its corresponding confidence score.

The confidence scores for each stem were graphically superposed across iterations. In the following graphs, the x-axis represents the iteration number, while the y-axis represents the confidence score. Different lines on the graph correspond to different stems, allowing for a comprehensive comparison.

We included the vocals stem to understand how an initial blank audio file changes through iterations. To assess the coherence of instruments across iterations relative to the original stem, we assigned a boolean value to each iteration based on whether the instrument name was included in the corresponding caption.

We then computed the Pearson correlation coefficient for each of the three iteration processes (one for each prompt: blues, classical, and reggae). The Pearson correlation coefficient, denoted as r , measures the linear correlation between two variables [17]. The coefficient ranges from -1 to 1, where $r = 1$ indicates a perfect positive correlation, $r = -1$ indicates a perfect negative correlation, and $r = 0$ indicates no correlation.

This approach allowed us to observe if the evolution of the iteration process showed any similarity depending on the initial genre.

4.1. Genre coherence - Prompt 1

The first prompt regarded a blues track, with specific emotional request.

Target Genre: Blues

Initial prompt: “A traditional blues track featuring a slow, soulful tempo. The song should be built around a 12-bar blues structure with prominent electric guitar riffs and a harmonica. Background should include a steady bass line and a minimalist drum beat to enhance the melancholic yet resilient spirit of blues.”

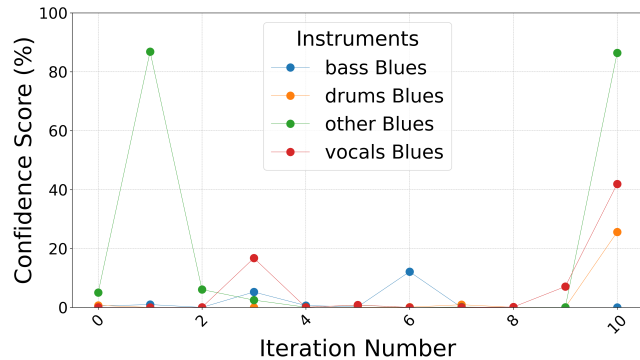


Figure 2: Prompt1 - Genre coherence assessment across all of the stems (bass, drums, other, and vocals) over ten iterations

4.2. Genre coherence - Prompt 2

The second prompt was crafted with consideration of music history, incorporating detailed descriptions of execution.

Target Genre: Classical

Initial prompt: “A classical music piece that embodies the elegance and complexity of the Baroque period. The composition should involve a full orchestra with a focus on string instruments like violins, violas, and cellos, complemented by harpsichords and woodwinds. Add intricate melodies and counterpoint techniques that highlight the skills of individual musicians.”

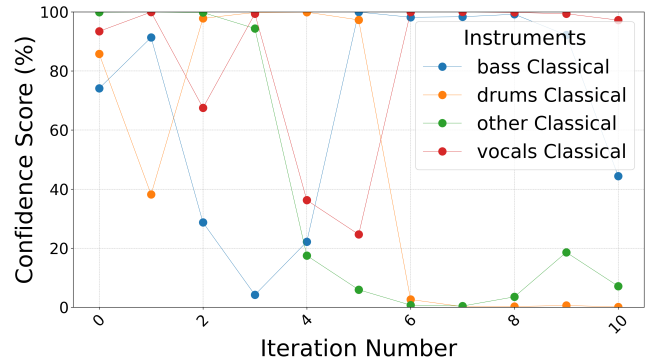


Figure 3: Prompt2 - Genre coherence assessment across all of the stems (bass, drums, other, and vocals) over ten iterations

4.3. Genre coherence - Prompt 3

The final prompt was designed to feature its characteristic instruments and included a specific description of its origin, aiming to test the model’s ability to associate the genre with its roots.

Target Genre: Reggae

Initial prompt: “A reggae track with a relaxed and upbeat rhythm characteristic of traditional reggae music. Use steady drum patterns, a bass guitar playing on the offbeat, and rhythmic guitar strumming. Include elements like the melodica or steel drums to give it an authentic island sound.”

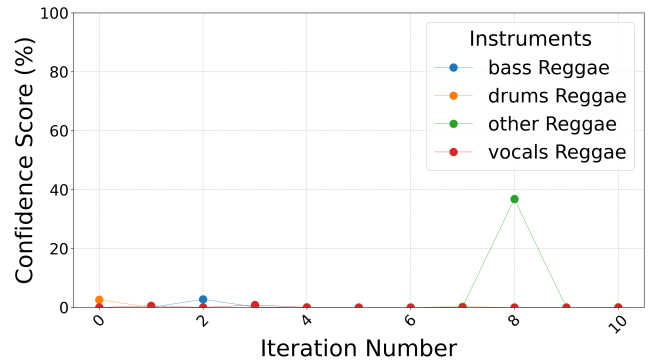


Figure 4: Prompt3 - Genre coherence assessment across all of the stems (bass, drums, other, and vocals) over ten iterations

4.4. Instruments coherence over iterations

Starting from the observation of the bass, drums, and other-instruments iterations for all three prompts, we constructed a correlation matrix, as shown in the following figure:

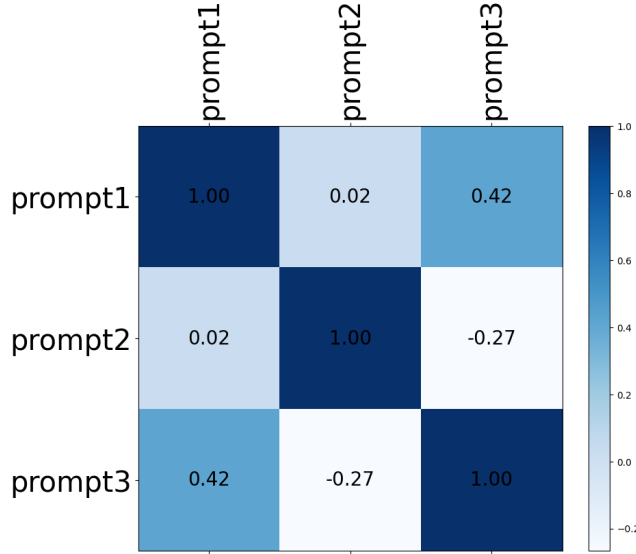


Figure 5: Pearson correlation matrix for the iterative processes

Prompt 1 (Blues) and Prompt 2 (Classical): The correlation coefficient between the Blues and Classical prompts is relatively low, indicating a weak positive correlation ($r = 0.02$). This weak correlation can be attributed to the significant differences in the structure and style of Blues and Classical music.

Prompt 1 (Blues) and Prompt 3 (Reggae): The correlation coefficient between the Blues and Reggae prompts shows a moderate positive correlation ($r = 0.42$). Blues and Reggae share some stylistic similarities, such as their emphasis on rhythm and the use of certain instrumental techniques.

Prompt 2 (Classical) and Prompt 3 (Reggae): The correlation coefficient between the Classical and Reggae prompts is negative, indicating an inverse relationship ($r = -0.27$). This negative correlation can be explained by the stark contrasts between the two genres.

5. DISCUSSION

The results of our study provide significant insights into the challenges and potentials of AI in music generation and audio captioning, especially concerning maintaining genre coherence across iterations. This section explores several key themes and poses questions for further exploration.

During our tests, we observed a high confidence and appropriate genre for all of the three initial prompts, despite the first being erroneously recognized as pop, although the feeling and emotional content were of a blues nature. The iterative processes initiated from the stems (i.e., bass, drums, other, vocals) exhibited very different behaviors among the prompts.

We observed that for the first and third prompts (i.e., blues and reggae), as you can see in Fig.2 and 4, the iterative genre coherence

was very low, with the exception of a slight increase for the first prompt in the last two iterations. In contrast, the second prompt (i.e., classical), see Fig. 3, showed much more iterative coherence. We believe this is due to the almost exclusively melodic nature of the initial prompt, influenced by the unsupervised melody conditioning of MusicGen.

We observed that audio captions, especially for some blank stems (e.g., bass and drums of the classical prompt number 2), were completely uncorrelated with the audio content, as if LP-Music Caps was obliged to write in an uncorrelated manner with respect to the music provided as input. This phenomenon, referred to as hallucination, occurs when the model generates text that, while confident in appearance, is not grounded in the actual input data. Hallucination makes it difficult to discern the accuracy of the generated text without a reference point or ground truth [11].

Notably, the iterations of the vocals stem for prompt 2 are of particular interest. Despite the audio file being mostly empty, they were fully coherent with the classical genre, confirming the melodic conditioning of MusicGen [6].

The Pearson correlation coefficients for instrument coherence suggest that some prompts retain instrumental characteristics better than others. *What specific aspects of the initial prompts contribute to this consistency? Could a human person correctly recognize the original genre starting from a random stem?*

Certainly, a more precise caption (i.e. the prompt for the subsequent iteration), without forcing its descriptive part, would have led to greater coherence both in terms of genre and instrumentation.

We should also consider the precision and error of the Convolutional Neural Network (CNN) employed to recognize the genres in the iteration process. The accuracy of genre recognition by the CNN is critical, as any misclassification can significantly impact the coherence of subsequent iterations. Misclassification can propagate through the iterative process, compounding errors and leading to divergence from the original genre.

Additionally, incorporating a perceptive evaluation could be a further consideration. Human listeners could provide qualitative feedback on the genre coherence and the overall musicality of the generated audio. A perceptive evaluation could also help in assessing the presence and impact of hallucinations in audio captions. Human evaluators could judge the relevance and accuracy of the generated captions in relation to the actual audio content.

6. CONCLUSION

The project demonstrated the cyclic process of generating music from text and vice versa, emphasizing both the potential and the challenges inherent in iterative generative models. While our study demonstrates the potential of AI in music generation and audio captioning, it also highlights several challenges, particularly regarding maintaining genre coherence and addressing hallucinations in audio captions. Future research should focus on improving the precision of genre recognition and understanding the factors that contribute to coherence across iterations. Incorporating perceptive evaluations as a further consideration will provide a more holistic understanding of the system's performance, bridging the gap between technical metrics and human experience.

All prompts and generated audio can be viewed on the project's GitHub page[18].

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